VNC remote control

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This tutorial applies to the bundled JetPack 7.2.1 / Ubuntu 24.04 system. It shares the local Jetson desktop through `x11vnc`, and Windows connects with VNC Viewer.

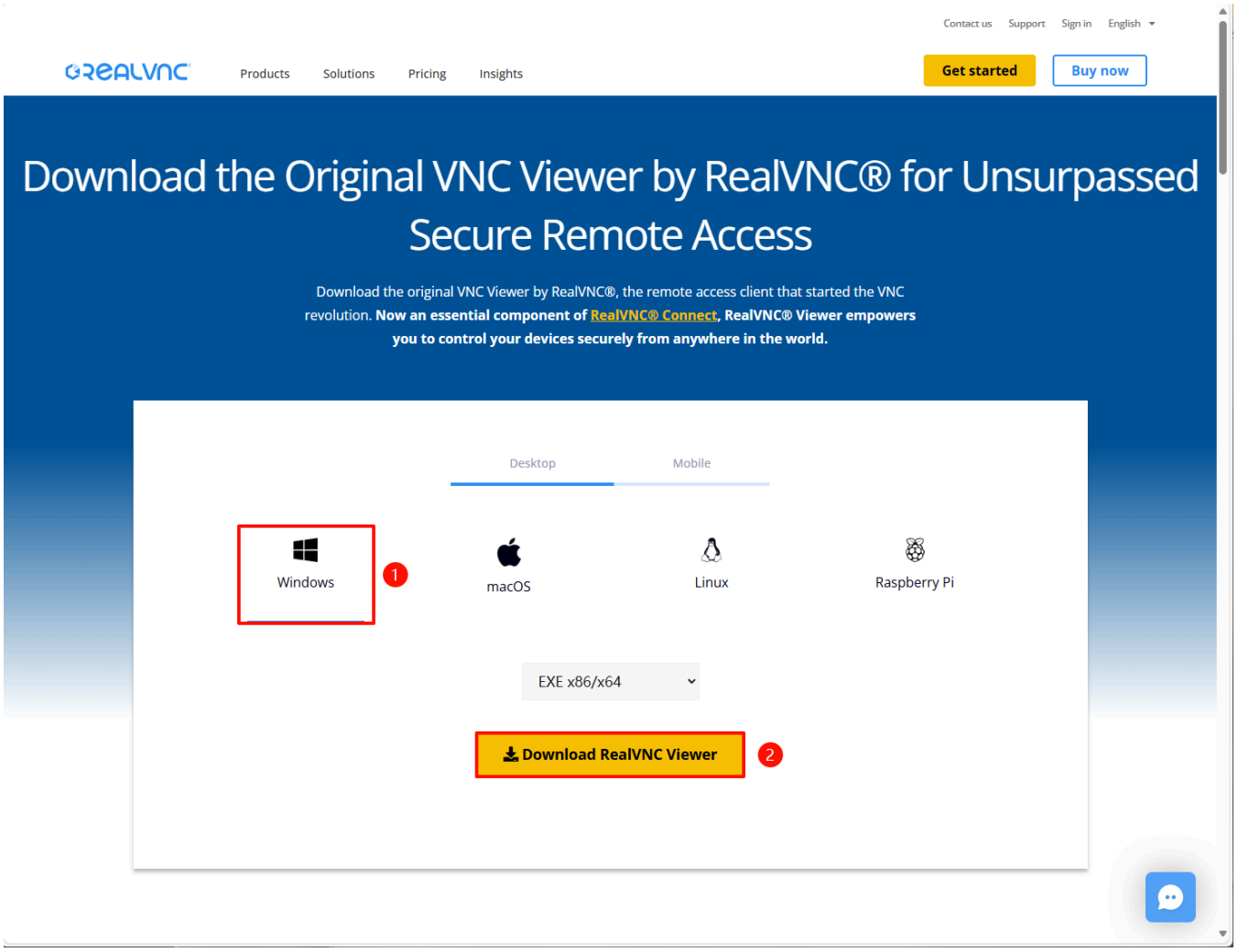
> The "Settings → System → Remote Desktop" page of Ubuntu 24.04 uses RDP and cannot act as the server side for VNC Viewer. This tutorial no longer uses the "Sharing → Remote Desktop → Legacy VNC" settings of Ubuntu 22.04. For RDP remote login, please refer to RDP remote control.

The Windows computer needs VNC Viewer installed in advance and has to be on the same LAN as the Jetson.

1. VNC Viewer

1.1. VNC download

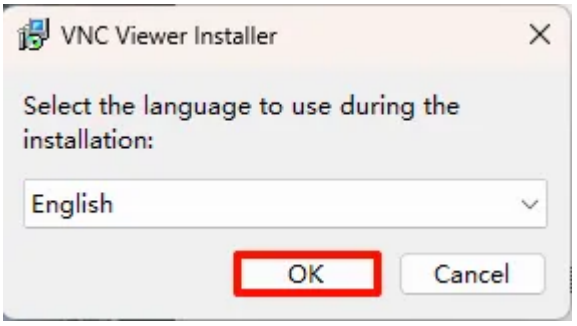
Official download address: https://www.realvnc.com/en/connect/download/viewer/

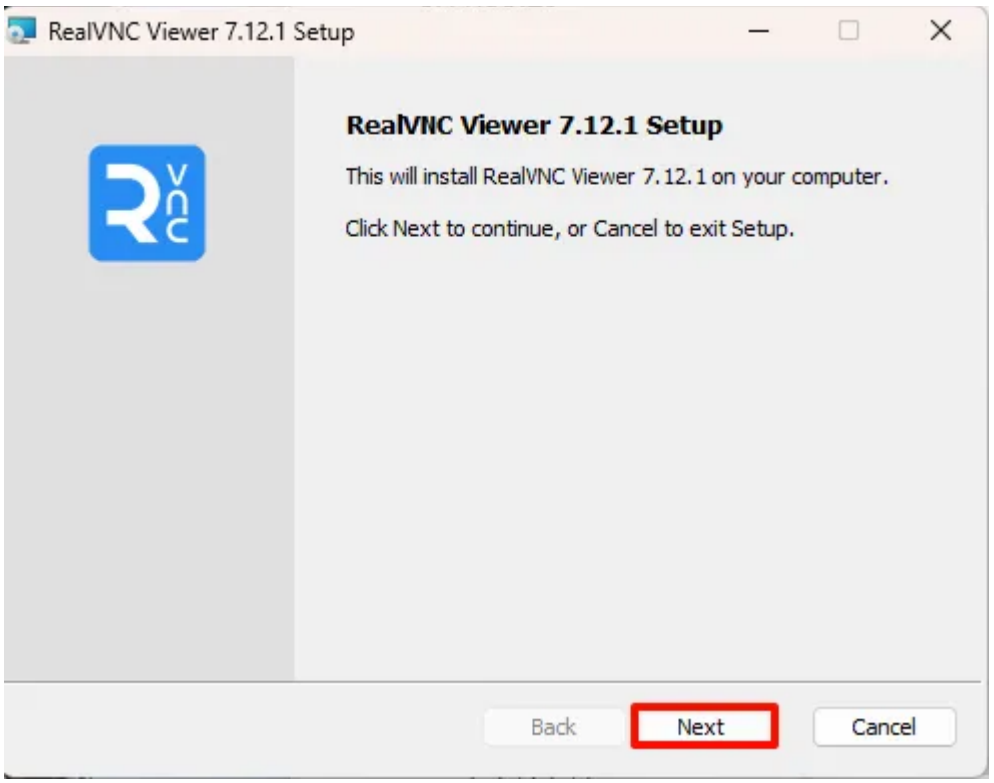


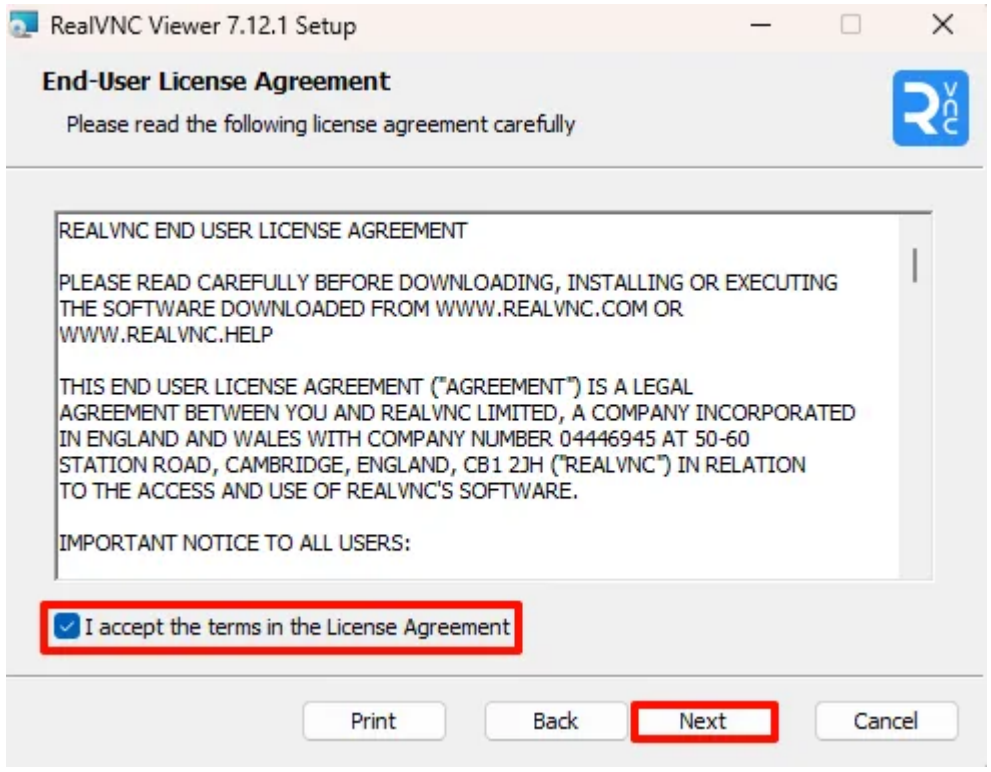
1.2. VNC installation

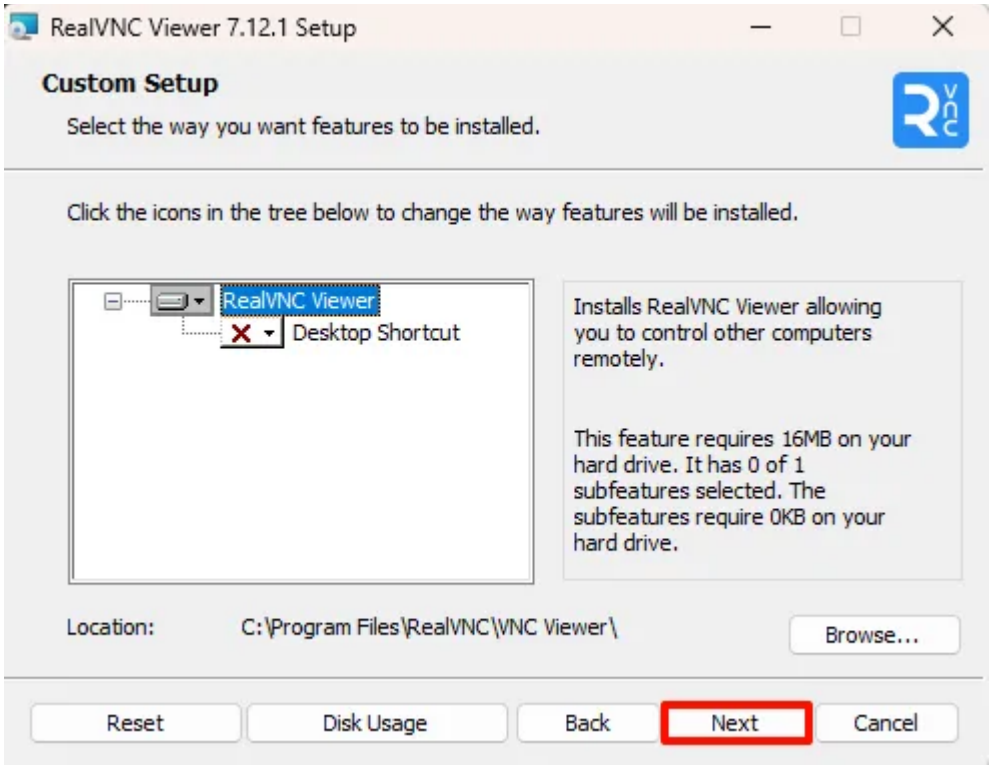
Run VNC-Viewer-xxx.exe as an administrator:

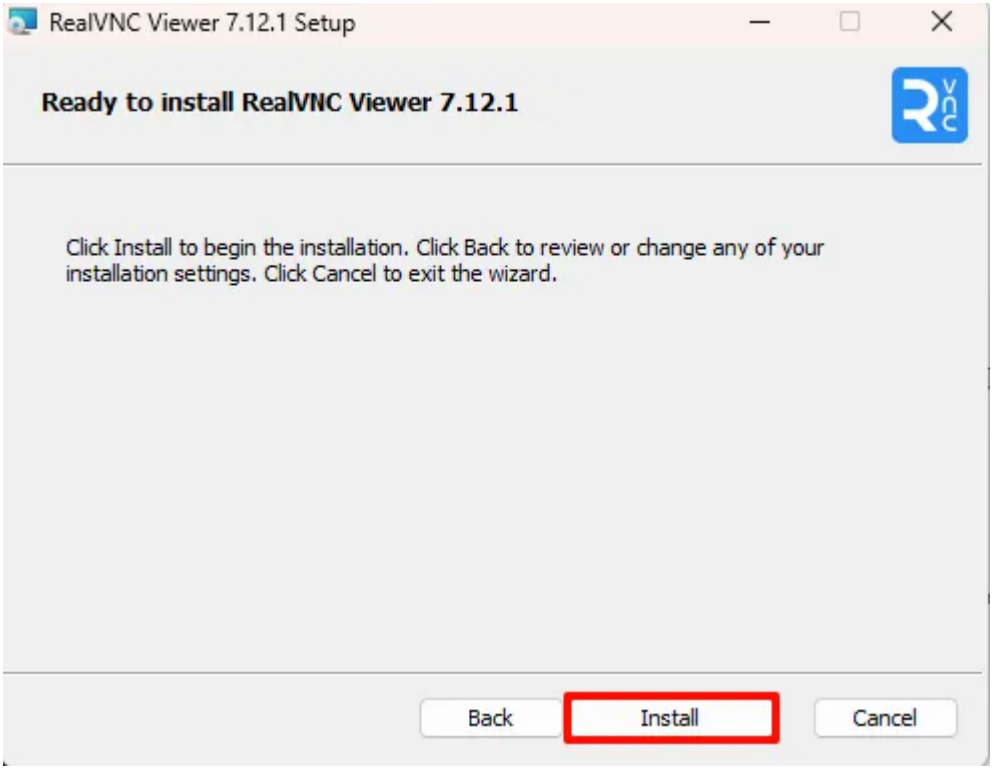


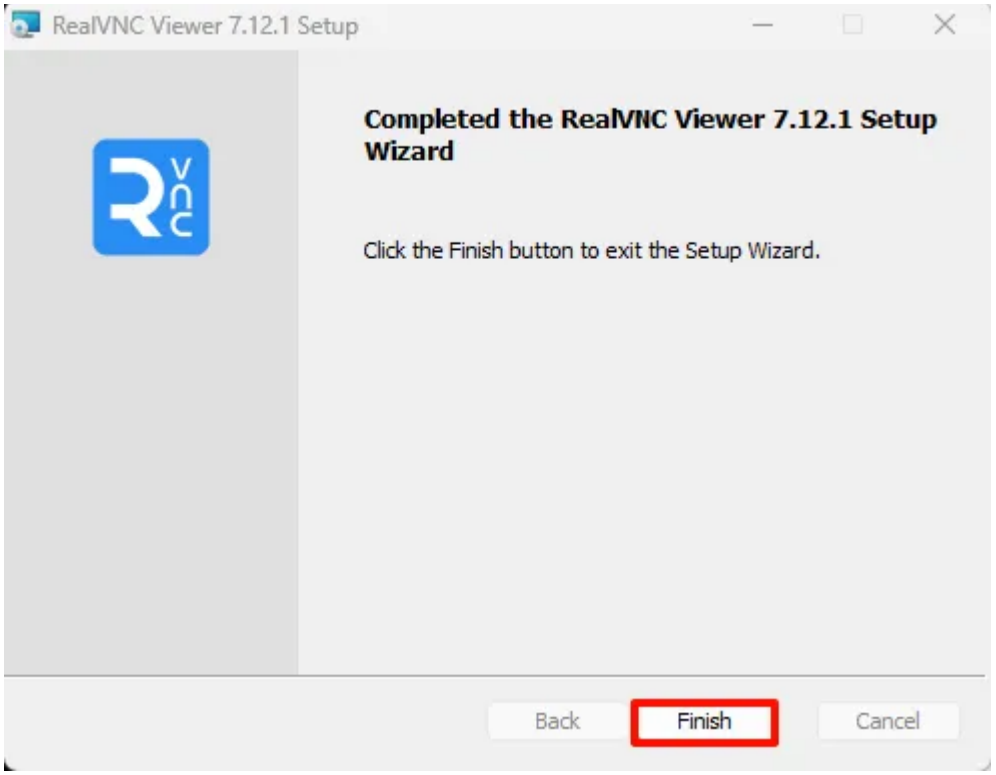



RealVNC Viewer 7.12.1 Setup
RealVNC Viewer 7.12.1 Setup
This will install RealVNC Viewer 7.12.1 on your computer.
Click Next to continue, or Cancel to exit Setup.
Back
Next
Cancel

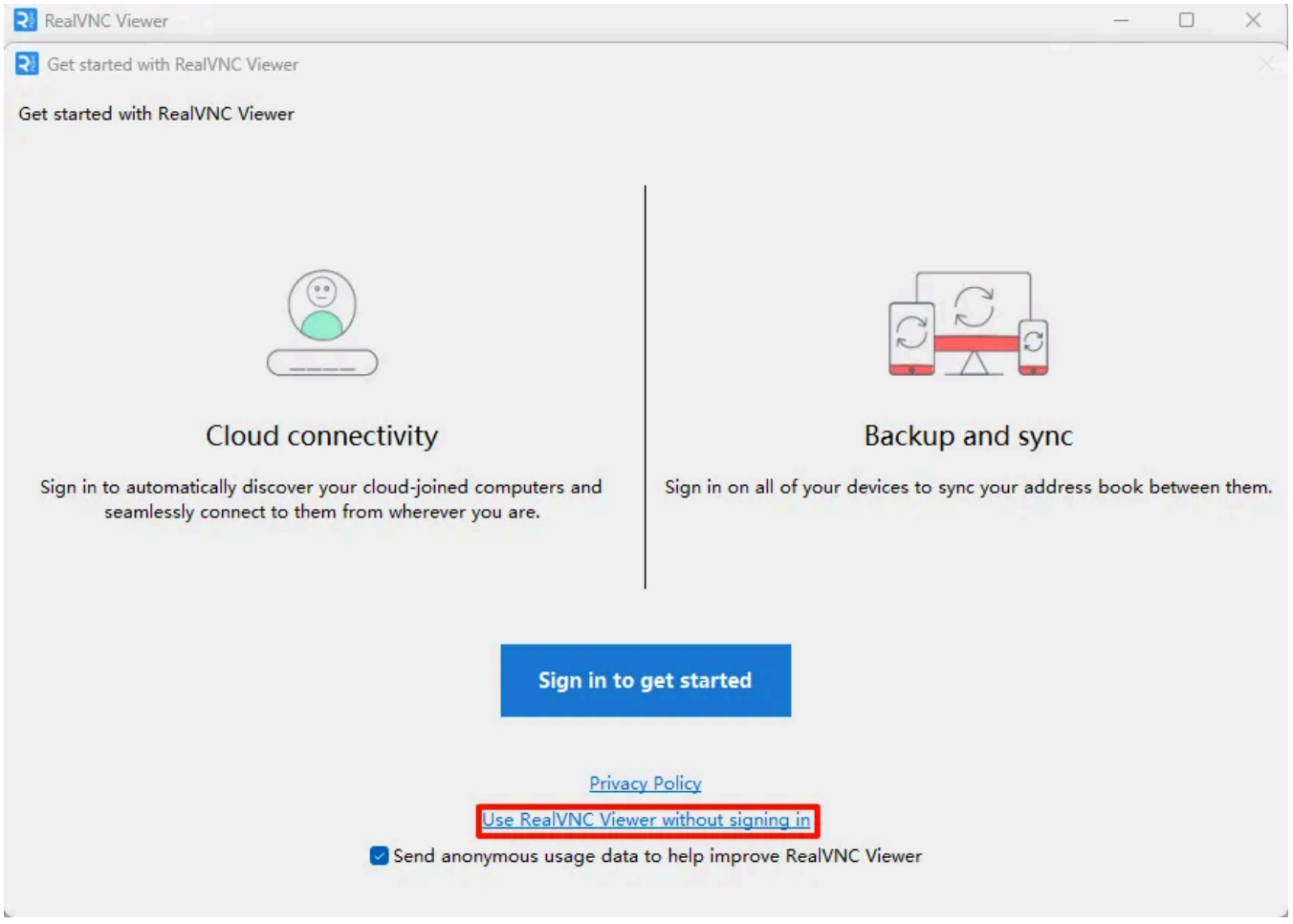

RealVNC Viewer 7.12.1 Setup
End-User License Agreement
Please read the following license agreement carefully
REALVNC END USER LICENSE AGREEMENT
PLEASE READ CAREFULLY BEFORE DOWNLOADING, INSTALLING OR EXECUTING THE SOFTWARE DOWNLOADED FROM WWW.REALVNC.COM OR WWW.REALVNC.HELP
THIS END USER LICENSE AGREEMENT ("AGREEMENT") IS A LEGAL AGREEMENT BETWEEN YOU AND REALVNC LIMITED, A COMPANY INCORPORATED IN ENGLAND AND WALES WITH COMPANY NUMBER 04446945 AT 50-60 STATION ROAD, CAMBRIDGE, ENGLAND, CB1 2JH ("REALVNC") IN RELATION TO THE ACCESS AND USE OF REALVNC'S SOFTWARE.
IMPORTANT NOTICE TO ALL USERS:
I accept the terms in the License Agreement
Print
Back
Next
Cancel

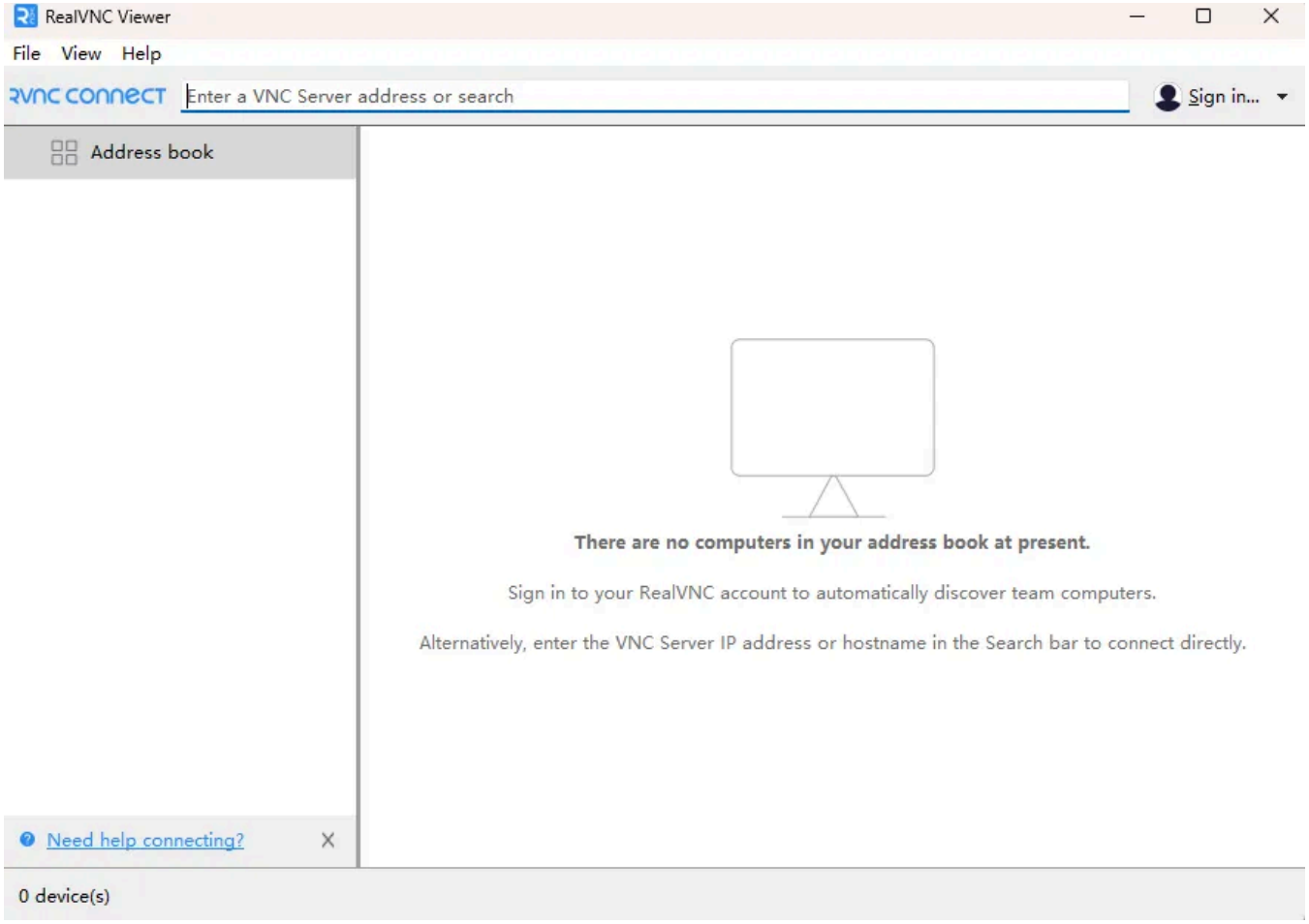

RealVNC Viewer 7.12.1 Setup
Custom Setup
Select the way you want features to be installed.
Click the icons in the tree below to change the way features will be installed.
RealVNC Viewer
Desktop Shortcut
Installs RealVNC Viewer allowing you to control other computers remotely.
This feature requires 16MB on your hard drive. It has 0 of 1 subfeatures selected. The subfeatures require 0KB on your hard drive.
Location:
C:\Program Files\RealVNC\VNC Viewer\
Browse...
Reset
Disk Usage
Back
Next
Cancel


RealVNC Viewer 7.12.1 Setup
Ready to install RealVNC Viewer 7.12.1
Click Install to begin the installation. Click Back to review or change any of your installation settings. Click Cancel to exit the wizard.
Back
Install
Cancel


RealVNC Viewer 7.12.1 Setup
Completed the RealVNC Viewer 7.12.1 Setup Wizard
Click the Finish button to exit the Setup Wizard.
Back
Finish
Cancel

1.3. VNC usage


RealVNC Viewer
Get started with RealVNC Viewer
Get started with RealVNC Viewer
Cloud connectivity
Sign in to automatically discover your cloud-joined computers and seamlessly connect to them from wherever you are.
Backup and sync
Sign in on all of your devices to sync your address book between them.
Sign in to get started
Privacy Policy
Use RealVNC Viewer without signing in
Send anonymous usage data to help improve RealVNC Viewer



2. System settings (Jetson)

2.1. Switch to VNC mode

The bundled system has `x11vnc` and a remote desktop mode switching script already configured. Run the following in the Jetson terminal or over SSH:

```
sudo switch-remote-desktop vnc
```

This command restores the local automatic login and the VNC shared desktop mode, and restarts `gdm3`. The switch interrupts the current graphical session, so save the work in your desktop programs before operating and confirm that a fallback connection such as SSH is available.

Check the current mode with:

```
sudo switch-remote-desktop status
```

> `switch-remote-desktop` is a script provided by the bundled system, not a command that ships with Ubuntu. If you flash the official pure NVIDIA system yourself, the script and the VNC service may be missing, and you have to install and configure `x11vnc` separately; you cannot simply copy the switching steps of the bundled system.

2.2. Confirm the desktop and the VNC service

The current VNC solution shares the local X11 desktop that is already logged in on the Jetson, usually the `:0` session on the display. It does not create an independent remote desktop; when nobody has logged in locally, when the graphical session has ended, or when Wayland is used, you cannot connect this way directly.

Check the service and the listening port on the Jetson:

```
systemctl status x11vnc.service --no-pager
ss -ltn | grep ':5900'
```

The service should show `active (running)` and listen on TCP `5900`. If the service is not running, look at the reason:

```
journalctl -u x11vnc.service -b -n 50 --no-pager
```

Provided the Jetson has already entered the local desktop, you can try restarting the service and then check the status again:

```
sudo systemctl restart x11vnc.service
```

Do not configure VNC through the "Media Sharing" page of Ubuntu; media sharing has nothing to do with VNC remote control. You also do not need to install the `allow-locked-remote-desktop` extension as the older tutorials describe, nor to clear the keyring password in "Passwords and Keys".

2.3. VNC password and connecting at boot

VNC Viewer uses the **VNC access password** configured in `x11vnc`, which is not necessarily the same as the system login password of the Jetson. Use the VNC password provided by the bundled system; do not substitute the credentials shown on the Ubuntu RDP page.

If you need to troubleshoot the password configuration or change the password, first look at the service startup parameters:

```
systemctl cat x11vnc.service
```

If the startup parameters contain `-rfbauth`, the path after it is the password file used by the service; when changing the password, operate on that actual file. Do not simply run `x11vnc -storepasswd` to generate another file and assume that the password of the running service has changed.

Connecting automatically at boot requires all of the following: the `x11vnc` service is enabled, the local X11 desktop is logged in, and TCP `5900` is reachable. Check whether the service is set to start at boot:

```
systemctl is-enabled x11vnc.service
```

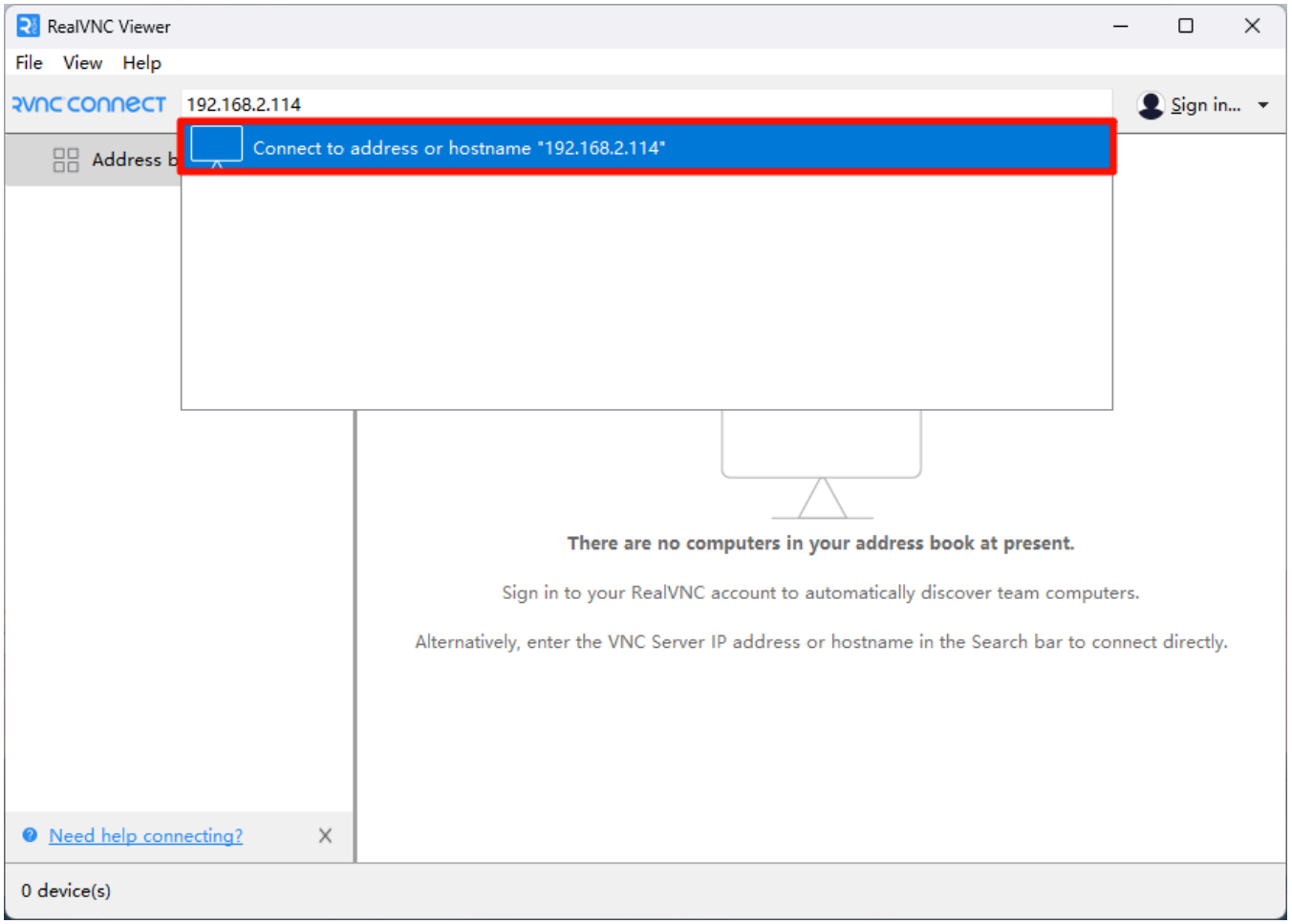
If it shows `disabled` and the service has been verified as working as described above, enable it:

```
sudo systemctl enable x11vnc.service
```

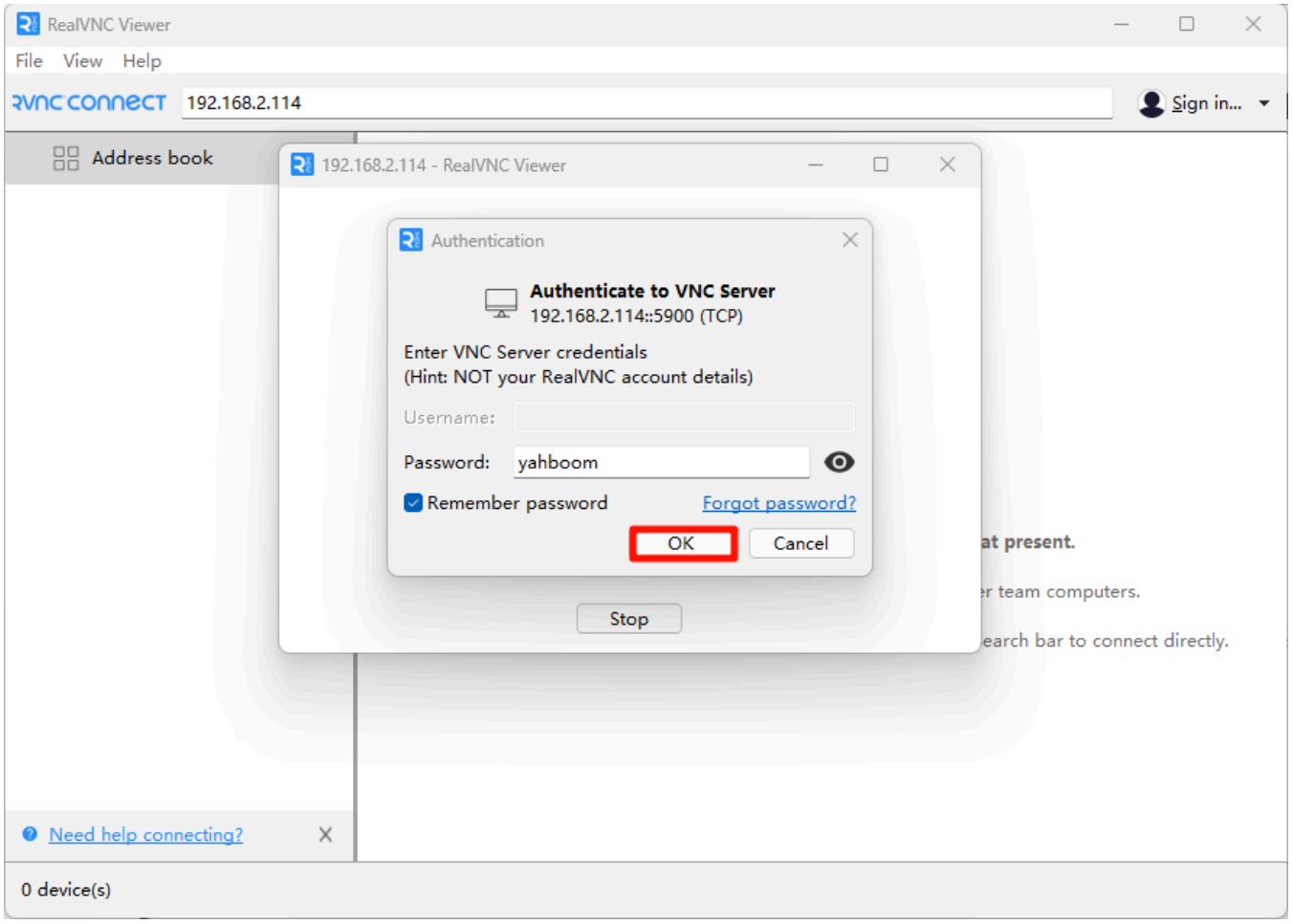
Enabling the service alone does not replace the local desktop login. The VNC mode of the bundled system depends on the local automatic login; switching to RDP mode turns the automatic login off, so you have to switch back to VNC mode to restore the original way of working. The behaviour of a connection after the screen is locked should be confirmed on the current image; the effect of the older GNOME extension does not apply to `x11vnc`.

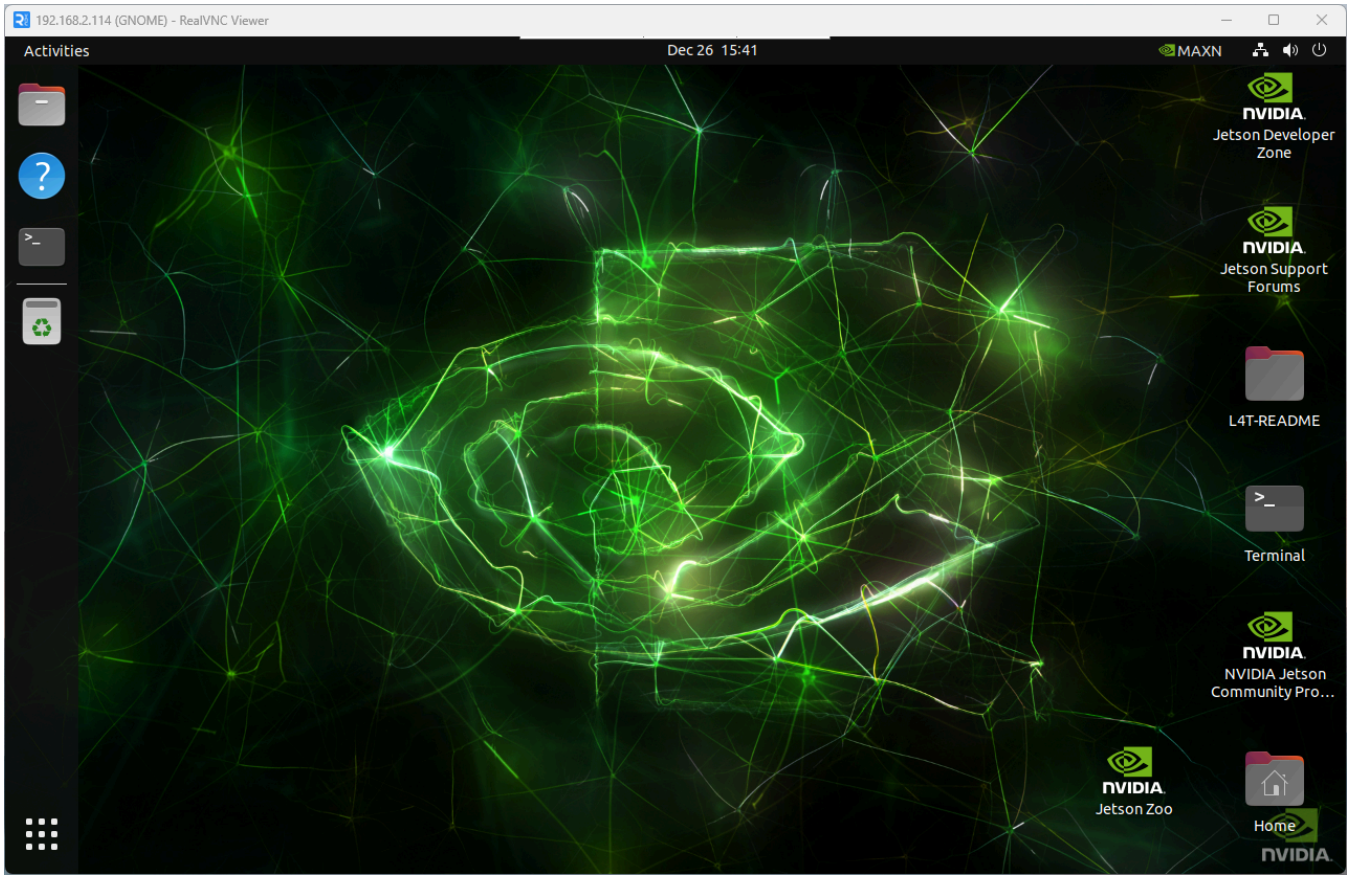
3. VNC remote control

Run `hostname -I` in the Jetson terminal to see the mainboard IP address. Enter that IP address in VNC Viewer; TCP `5900` is used by default, and you can also enter `IP:5900` (replace `IP` with the actual address).



Enter the VNC access password when prompted (see section 2.3); it is not the system login password. The screenshots below only illustrate the client connection flow; for the addresses and credentials in the old screenshots, follow the current device.


RealVNC Viewer
File View Help
192.168.2.114
Sign in...
Address book
192.168.2.114 - RealVNC Viewer
Authentication
Authenticate to VNC Server
192.168.2.114::5900 (TCP)
Enter VNC Server credentials
(Hint: NOT your RealVNC account details)
Username:
Password:
yahboom
Remember password
Forgot password?
OK
Cancel
Stop
at present.
r team computers.
earch bar to connect directly.
Need help connecting?
0 device(s)


192.168.2.114 (GNOME) - RealVNC Viewer
Activities
Dec 26 15:41
MAXN
NVIDIA Jetson Developer Zone
NVIDIA Jetson Support Forums
L4T-README
Terminal
NVIDIA NVIDIA Jetson Community Pro...
NVIDIA Jetson Zoo
Home
NVIDIA

Frequently asked questions

Cannot find the VNC switch for the remote desktop

The remote desktop that ships with Ubuntu 24.04 uses RDP. The "Settings → System → Remote Desktop" page cannot enable the `x11vnc` service used in this tutorial; on the bundled system, use the mode switching command in section 2.

`switch-remote-desktop: command not found` is reported

Check whether you are using the bundled system or a self-installed official pure system. The command is a bundled script and cannot be filled in by installing a GNOME extension. First check whether the `x11vnc` service has been installed and configured; if `systemctl status x11vnc.service` reports that the service does not exist, you have to configure the VNC server side separately.

Cannot connect, or a black screen after connecting

First confirm that the IP address of the Jetson is reachable from the computer, then check the service and the `5900` listener described in section 2.2. Confirm that you have switched back to VNC mode, that the local desktop of the Jetson is logged in, and that neither the network nor the firewall blocks `5900`. When there is no display, you also have to confirm that the system has actually created a shareable X11 desktop; VNC availability cannot be assumed just because SSH works.

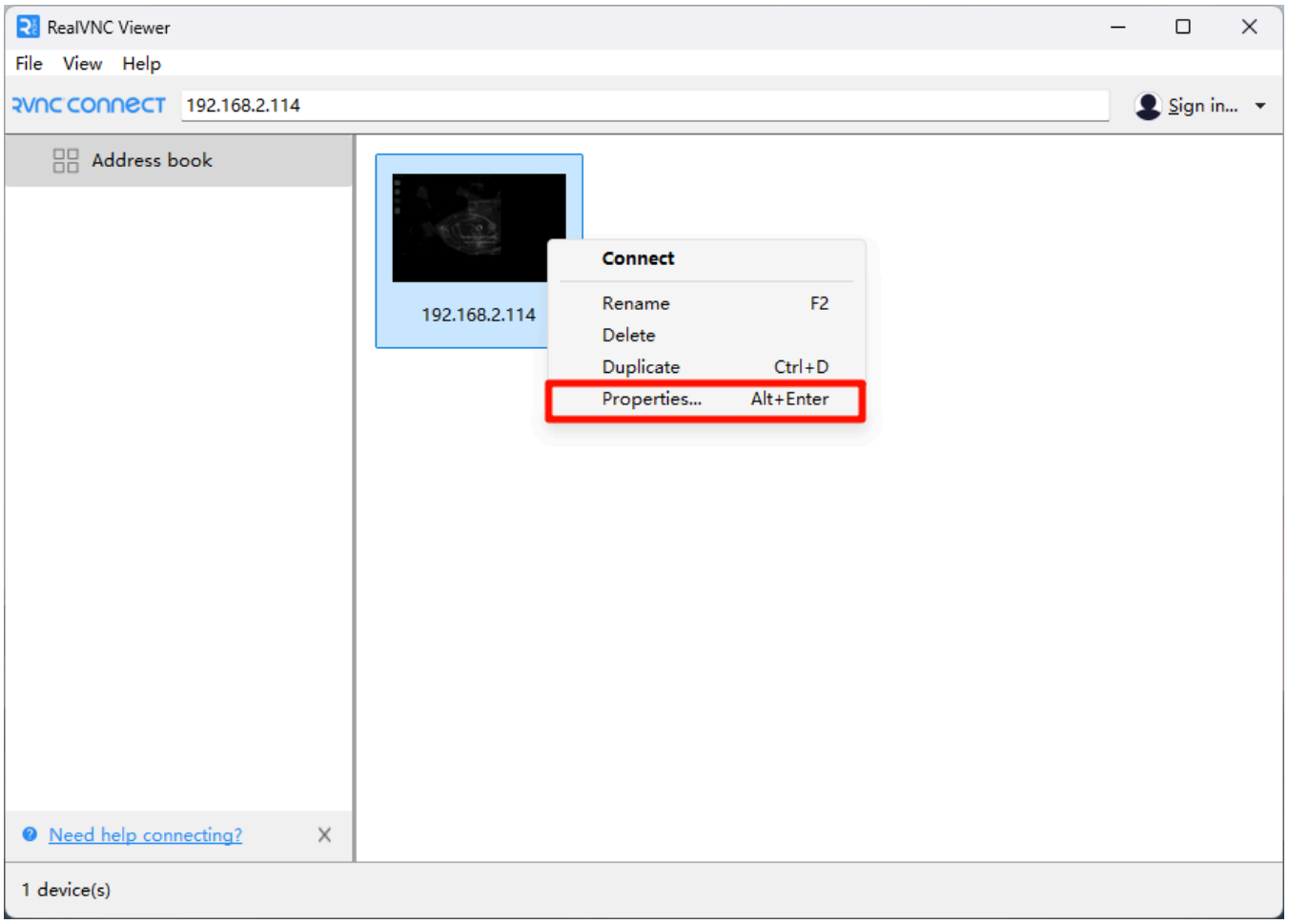
VNC remote display reconnection

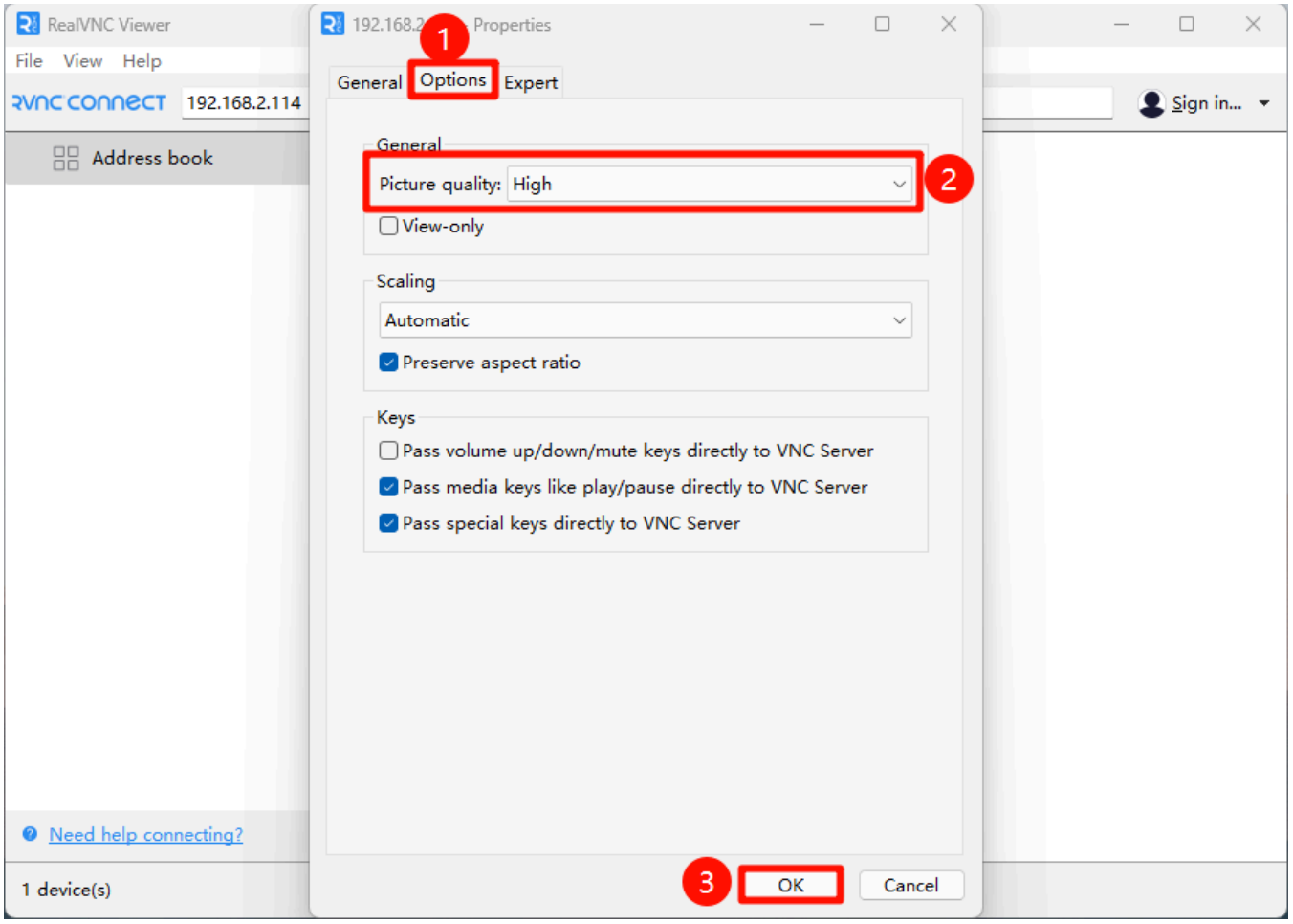
Reconnection phenomenon

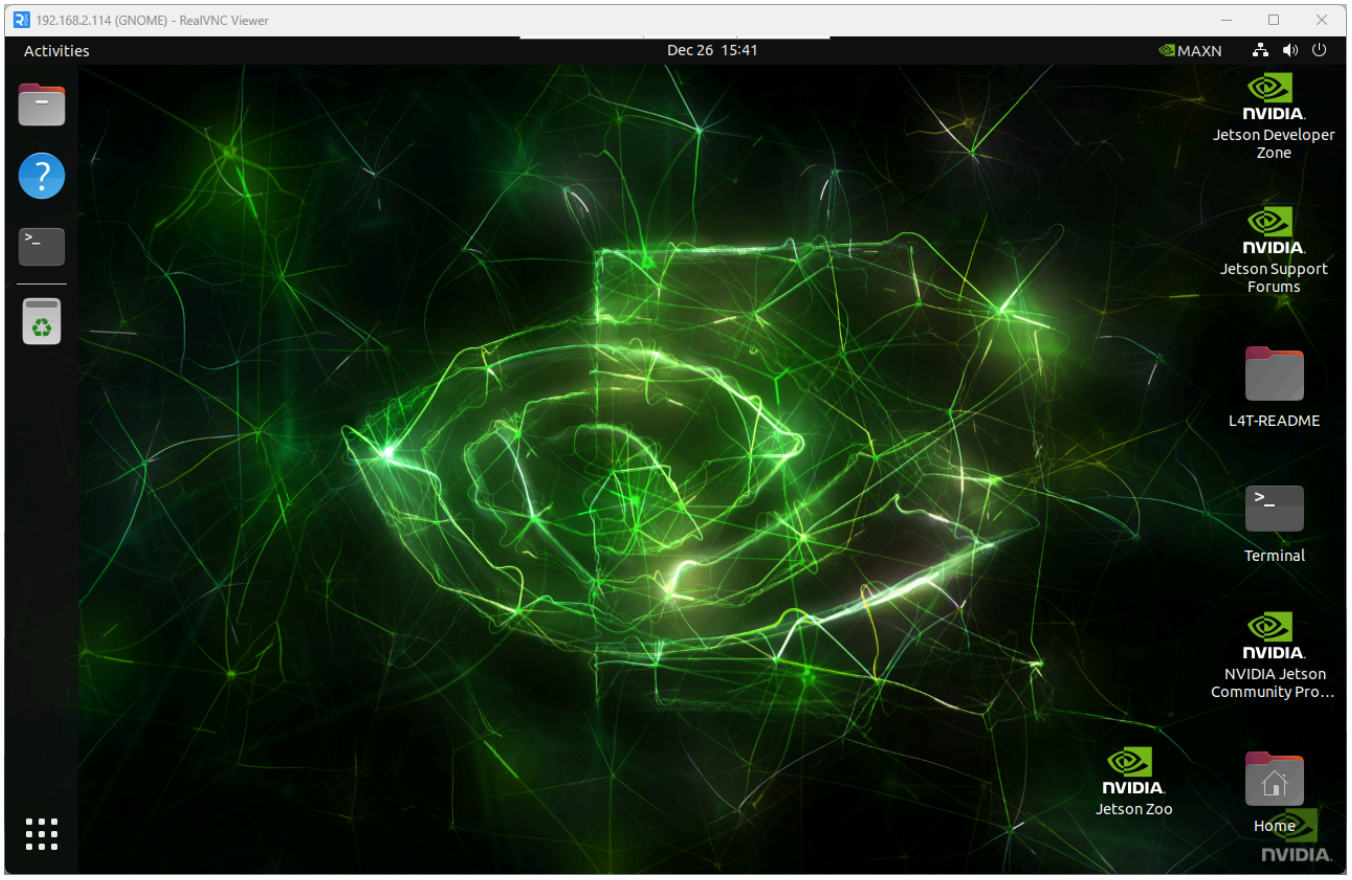


Solution

Modify the options of the corresponding remote device → Specify remote image quality




RealVNC Viewer
File View Help
VNC connect
192.168.2.114
Address book
Need help connecting?
1 device(s)
192.168.2 Properties
General
Options
Expert
1
General
Picture quality: High
2
View-only
Scaling
Automatic
Preserve aspect ratio
Keys
Pass volume up/down/mute keys directly to VNC Server
Pass media keys like play/pause directly to VNC Server
Pass special keys directly to VNC Server
3
OK
Cancel
Sign in...


192.168.2.114 (GNOME) - RealVNC Viewer
Activities
Dec 26 15:41
MAXN
NVIDIA Jetson Developer Zone
NVIDIA Jetson Support Forums
L4T-README
Terminal
NVIDIA NVIDIA Jetson Community Pro...
NVIDIA Jetson Zoo
Home